

Week 4 · Student Notes

AI531C Digital Image Analysis · Tue 25 Aug 2026 · Unit II · Lectures 10

Everything taught in this week, in one document. The per-lecture handouts are identical in content — use whichever you prefer.

What is in this bundle

- Lecture 10 — The Image Degradation/Restoration Model and Noise Processes

This week at a glance

Sessions — Tue 25 Aug 2026.

Tutorial — T4 — Assessment Test I (Tue 25 Aug).

Pre-class preparation — PCM-4.

Released after the last class of the week: Tue 25 Aug 2026, 18:00.

Short week: Wed 26 Aug is Milad-un-Nabi.

How to use this bundle — read the lecture section on the evening of the class while the argument is fresh, attempt the check-your-understanding questions before the tutorial, and leave the practice questions until you can do them unaided. No answer keys are printed: the reasoning is worked through in the tutorial.

Lecture 10 · The Image Degradation/Restoration Model and Noise Processes

AI531C Digital Image Analysis · Unit II · CO2 · M.Tech. Artificial Intelligence, Department of CSE, DTU

Unit II opens: everything that went wrong between the scene and your array — named, modelled, and measured

What you should already know

This lecture continues from Lecture 9, “Fourier-Domain Image Analysis and Multi-Scale Representation”. Revise that material first — today's arguments assume it.

Reading in the course notes: Comprehensive Notes L10; Vol. I ch. 10.

Learning outcomes

After studying this lecture you should be able to:

The degradation model

Problem: relate the observed image g to the true scene f . Assume the degradation operator is linear and shift-invariant (L8's two properties), and noise enters additively after it.

$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y) \quad G(u,v) = H(u,v) \cdot F(u,v) + N(u,v)$$

What each symbol means

- f — true image; g — observed image h — degradation PSF (blur kernel); H — its spectrum (OTF)
- η — random noise, PDF $p(\eta)$; N — its spectrum $*$ — convolution (hence the frequency product, L9)

How to read this

What it means — Blur mixes neighbouring truth; noise adds lies on top. Restoration must un-mix without amplifying the lies — the whole tension of L12. Every symbol here is measurable or estimable: h from a point source or edge; η from a flat region.

Core concepts

The five noise processes and where they come from

Gaussian — the electronics — Thermal and read noise; many small independent contributions → central limit theorem. Additive, signal-independent. $p(\eta) = N(0, \sigma^2)$.

Impulse (salt & pepper) — the failures — Dead/hot pixels, bit errors, transmission dropouts. A fraction of pixels replaced by extremes; the rest untouched.

Poisson — the photons — Light itself arrives in counts; variance equals the mean.

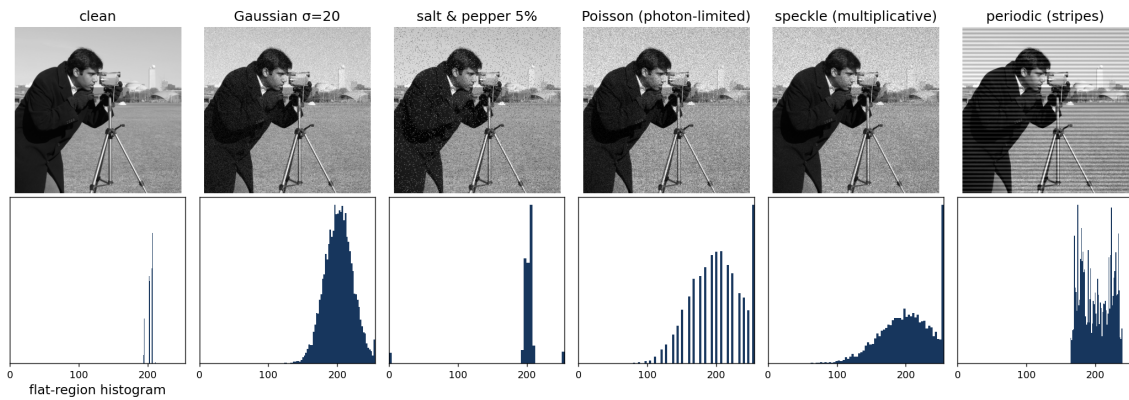
Signal-DEPENDENT: dark regions are relatively noisier — the low-light lane's core problem.

Speckle — coherent imaging — Ultrasound, SAR, laser: interference of coherent waves.

MULTIPLICATIVE: $g = f \cdot (1 + n)$.

Periodic — the environment — Mains hum, screen refresh, vibration. Deterministic → spectral spikes → notch (L9).

Figure 1: Reading noise from a flat region



camera() sky region, 100×100 px, under each process. Generated by L10_noise_processes.py.

How to read this figure

What to notice — Gaussian: symmetric bell around the true value. Impulse: two extra spikes at 0 and 255. Speckle: widened AND skewed — width grew with brightness.

Why it happens — A flat region has almost no scene variance — what remains IS the noise PDF, sampled 10 000 times.

What it means — The histogram of a flat patch is a noise-identification instrument.

Why it matters — Every denoising decision in L11 starts with this one look.

Worked example

Given — A flat sky patch measures $s_{\text{obs}} = 20.3$ in the corrupted image; the same patch in a clean reference measures $s_{\text{scene}} = 4.2$ (residual texture). Noise is additive and independent of the scene.

Find — The noise standard deviation σ ; and the fraction of pixels shifted by more than 2σ .

Formula — Independence $\Rightarrow$ variances add: $s_{\text{obs}}^2 = s_{\text{scene}}^2 + \sigma^2$; $P(|\eta| > 2\sigma) = 2(1 - \Phi(2))$

Solution

Substitution — $\sigma^2 = 20.3^2 - 4.2^2 = 412.1 - 17.6 = 394.5$

Calculation — $\sigma = \sqrt{394.5} = 19.9$; $P(|\eta| > 2\sigma) = 2(1 - 0.9772)$ ANSWER: $\sigma \approx 19.9$ (true value 20); $\approx 4.6\%$ of pixels beyond $\pm 2\sigma$

What it means — Variances add, standard deviations do not — subtracting 4.2 from 20.3 directly would be wrong. The 4.6% tail is why ' $\sigma = 20$ ' still means thousands of badly-shifted pixels in a megapixel image.

The code behind the results

```
noisy = img + rng.normal(0, 20, img.shape)
m = rng.random(img.shape)
# impulse img[m < 0.025] = 0; img[m > 0.975] = 255 photon = rng.poisson(img / 8.0) * 8.0
speckle = img * (1 + rng.normal(0, .2, img.shape))
```

Parameters used — $\sigma=20$; density 5%; photon scale 8; speckle $\sigma=0.2$. Full program:

L10_noise_processes.py

What it means — Simulation gives ground truth — the only way PSNR in L11 can exist.

Something to think about — Synthetic noise mismatches the camera's? The trained denoiser fails on real images (L30).

The noise family — reference table

Noise	Origin	Model	Histogram clue	First-choice cure (L11/L9)
Gaussian	Electronics (CLT)	additive, signal-indep.	symmetric bell	Gaussian/bilateral filter
Impulse	Dead pixels, bit errors	replacement, sparse	spikes at 0 & 255	median filter
Poisson	Photon counting	signal-DEP., var = mean	width grows with brightness	longer exposure; variance-stabilise
Speckle	Coherent interference	multiplicative	skewed, scales with signal	median/adaptive; log-domain
Periodic	Interference sources	deterministic sinusoid	(see spectrum, not histogram)	notch filter (L9)

Key takeaway: identify before you filter — L11 turns this table's last column into evidence with PSNR.

Core concepts (continued): Where the additive model itself fails

Poisson breaks additivity — Noise variance equals local brightness — one global σ describes nothing. A denoiser tuned on bright regions under-cleans shadows (and vice versa). Low-light work (L33) lives inside this failure.

Speckle breaks it multiplicatively — $g = f \cdot (1+n)$: the damage scales with the signal. Standard trick: work in the log domain ($\ln g = \ln f + \ln(1+n)$) — the homomorphic move from L9, reused.

Estimation from the wrong region — Measure σ on texture instead of a flat patch and scene variance inflates the estimate (the numerical's s_{scene} term, ignored). Automated flat-region finders exist; a human eye and a histogram remain the honest baseline.

Where this is used — your project

Denoising lane (and PCM-5 for everyone) — your dataset card must NAME the noise model, its parameters, and the evidence (flat-patch histogram) — 'noisy images' is not a specification;

Low-light lane — state whether your degradation is Poisson-dominated (real low light) or Gaussian (synthetic dimming) — they need different methods and different claims;

All lanes — simulated degradations are legitimate ONLY when documented and matched to the target domain — the T5 studio makes this an explicit graded step.

Try this yourself — Find one genuinely flat region in your project imagery; histogram it; write down what you can and cannot conclude.

Summary

- Restoration begins with a damage model: $g = h * f + \eta$ — every term measurable. Noise families have physical origins and histogram fingerprints — identify before filtering. Additivity is an assumption: Poisson and speckle violate it in exactly the ways real projects meet.

Key formulae

$g = h * f + \eta \Leftrightarrow G = H \cdot F + N$ Flat patch: $\sigma^2 = s_{\text{obs}}^2 - s_{\text{scene}}^2$ Poisson: $\text{var} = \text{mean}$; speckle: $g = f \cdot (1+n)$ $P(|\eta| > 2\sigma) \approx 4.6\%$ (Gaussian)

What comes next

Tutorial T4 — today 2:00–3:00 PM — Assessment Test I (L1–L10: conceptual 2.0 + numerical 2.0 + interpretation 1.0 marks), then the Fourier studio: magnitude/phase visualisation and notch filtering on periodic noise.

PCM-5 (releases Fri 28 Aug, due Mon 31 Aug, ~35 min) — Noise detective: classify five noisy crops, choose a plausible denoising method for each with one reason and one expected side effect.

Logistics — Next lecture L11 (Tue 1 Sep): spatial denoising with PSNR evidence. T5 (1 Sep): noise & restoration studio — PCM-5 feeds it directly.

Check your understanding

Work these before the tutorial. They are the same questions used in class; the reasoning is discussed there, so attempt them first.

1. Noise detective (PCM-5 preview)

For each scenario, name the dominant noise family and ONE clue you would check: A. Night-sky phone photo at high ISO. B. Image received over a faulty serial link. C. Ultrasound scan of the liver. D. Microscope video with faint rolling bands.

Practice questions

Attempt these unaided, then compare with the discussion in your tutorial session.

Q1 (4 marks, remember/understand): Write the degradation model in both domains, defining every symbol, and state its two structural assumptions.

Q2 (6 marks, apply): A flat region reads std 15.2 in a corrupted image and 3.5 in a clean reference. Estimate σ , and the percentage of pixels perturbed beyond $\pm\sigma$ (Gaussian noise).

Q3 (10 marks, analyse): A team denoises real night-time photographs with a filter tuned for additive Gaussian noise of fixed σ and finds shadows still noisy while bright areas blur. Diagnose using the correct noise model and propose a two-step remedy.

Further reading

GW4 Ch.5 Secs.5.1-5.3; Szeliski Ch.3 regularised perspective

Course notes: Comprehensive Notes L10; Vol. I ch. 10.

Runnable code for every figure and number in this lecture: 06_CODE_AND_DEMOS (see the file names quoted beside each figure).

AI531C Digital Image Analysis · student handout · every figure and numerical result in this document is the output of course code you can run yourself.